Aakash
Medical | IIT-JEE | Foundations

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MM : 720 NCERT Booster Test Series for NEET (RM)-2024-25_T01D **Time : 200 Min.**

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification, Morphology of Flowering Plants
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation, Excretory Products & their Elimination, Locomotion & Movement

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

SECTION-A

1. A body is projected vertically upward with speed 20 ms^{-1}. The displacement of body from $t = 0$ s to $t = 4$ s is [g = 10 m s^{-2}]

(1) 20 m

(2) 40 m

(3) Zero

(4) 10 m

2. Dimensional formula of pressure is

(1) $[MLT^{-2}]$

(2) $[ML^{-1}T^{-2}]$

(3) $[M^{-1}L^{2}T^{-2}]$

(4) $[ML^{2}T^{-2}]$

3. If mass (M), acceleration (A) and velocity (V) are taken as base quantities, then dimensional formula of energy is

(1) $[MV^{-2}A^{4}]$

(2) $[MV^{2}A^{0}]$

(3) $[M^{-1}V^{3}A^{-1}]$

(4) [MVA]

4. A train of 360 m length is going along east with 50 m/s. A bird flies at a speed 10 m/s towards west parallel to the track. The time taken by the bird to cross the train is

(1) 10 s

(2) 8 s

(3) 6 s

(4) 4 s

5. The velocity of an object moving in a straight line changes from 20 m/s to 144 km/h in 10 s. Average acceleration of the object in this time interval is

(1) 12.4 m/s^2

(2) 16.4 m/s^2

(3) 2 m/s^2

(4) 10.2 m/s^2

6. The position of a particle moving along *y*-axis is given as $y = (5t^2 - 10t + 2)$ m. The position of particle when its velocity is zero, is

(1) Zero

(2) –3 m

(3) +3 m

(4) +1 m

7. If $k\hat{i} + \frac{1}{2}\hat{j}$ is a unit vector, then *k* can be equal to

(1) $\frac{\sqrt{3}}{2}$

(2) $\frac{1}{2}$

(3) 1

(4) 2

8. To the driver of a car *A* moving with velocity $\vec{v}_A = \left(3\hat{i} - 4\hat{j}\right)$ m/s, a second car *B* appears to have a velocity $\left(5\hat{i} + 12\hat{j}\right)$ m/s. The true velocity of the car *B* is

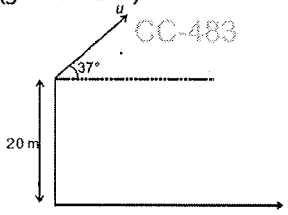
(1) $\left(-2\hat{i} - 16\hat{j}\right)$ m/s

(2) $\left(8\hat{i} + 8\hat{j}\right)$ m/s

(3) $\left(-8\hat{i} + 8\hat{j}\right)$ m/s

(4) $\left(2\hat{i} + 16\hat{j}\right)$ m/s

9. A body is projected with velocity $u = 50$ ms^{-1} at an angle of 37° to the horizontal from the top of a tower of height 20 m. The vertical component of velocity of body just before hitting the ground is ($g = 10$ ms^{-2})



(1) $10\sqrt{13}$ m/s

(2) $10\sqrt{5}$ m/s

(3) 40 m/s

(4) $10\sqrt{29}$ m/s

10. Position of a particle can be expressed as $\vec{r} = \left(9t^2\hat{i} + 3t\hat{j}\right)$ m. The magnitude of acceleration of the particle is (*t* is in second)

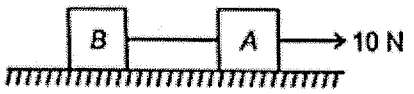
(1) 18 m/s^2

(2) 9 m/s^2

(3) Zero

(4) $9\sqrt{5}\text{m/s}^2$

11. Two blocks A and B having masses 3 Kg and 2 Kg respectively are connected by a massless string and placed on a smooth horizontal surface as shown in the figure. If a force of 10 N is applied on block A, then tension in the string connecting two blocks will be



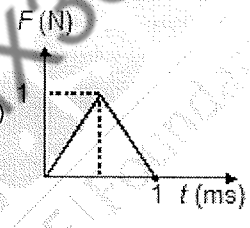
(1) 10 N

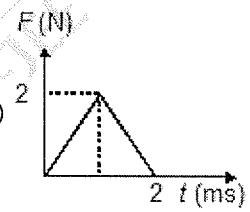
(2) 4 N

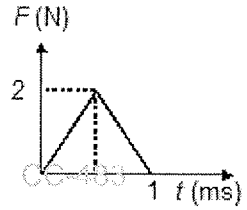
(3) 6 N

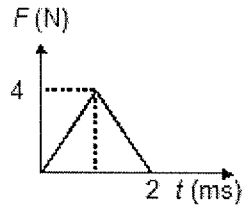
(4) 2 N

12. Which of the following force time graph, gives maximum impulse?

(1) 

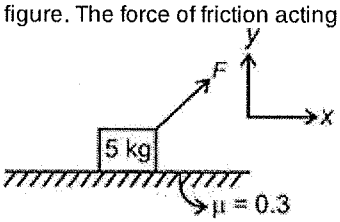
(2) 

(3) 

(4) 

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13. A force $\vec{F} = \left(\hat{i} + 4\hat{j}\right)$ N acts on a block as shown in the figure. The force of friction acting on the block is



(1) $-\hat{i}$ N

(2) $-1.8\hat{i}$ N

(3) $-2.4\hat{i}$ N

(4) $-3\hat{i}$ N

14. In uniform circular motion, which of the following quantities is zero?

(1) Change in kinetic energy

(2) Magnitude of change in velocity

(3) Change in magnitude of velocity

(4) Both (1) & (3)

15. A rope is wound around a uniform solid cylinder of mass 6 kg and radius 80 cm. The rope is pulled with a force F such that angular acceleration of the cylinder is 10 rad/s^2. The value of F is

(1) 61 N

(2) 10 N

(3) 12 N

(4) 24 N

16. The rotational inertia of a disc about its axis is 0.70 kg m^2. When a 2.0 kg mass is added to its rim, 0.40 m from the axis, the rotational inertia becomes

(1) 0.54 kg m^2

(2) 0.80 kg m^2

(3) 1.02 kg m^2

(4) 2.02 kg m^2

17. Potential energy function is defined for

(1) Conservative forces

(2) Non-conservative forces

(3) Air drag

(4) Both (1) and (2)

18. A block is pushed through 2 m across a surface offering 25 N resistance. The work done by resisting force is

(1) 50 J

(2) 25 J

(3) – 50 J

(4) – 25 J

19. A force F = 20 N acts on a body of mass 5 kg at rest, then power of force at time 2 s will be

(1) 40 W

(2) 80 W

(3) 160 W

(4) 320 W

20. In a perfectly elastic collision between two particles which of the following is conserved

(1) Kinetic energy of each particle

(2) Linear momentum of the system

(3) Speed of each particle

(4) Linear momentum of each particle

21. The angular momentum of a particle moving in a circular orbit with constant speed remains constant about

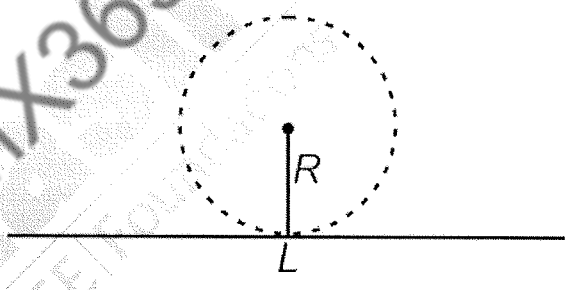
(1) Any point inside circle

(2) Any point outside circle

(3) Any point on the circumference

(4) The centre of circle

22. A uniform wire of length L is bent in the form of circle as shown in figure. The shift in the centre of mass is



(1) $\frac{L}{2\pi}$

(2) $\frac{L}{3\pi}$

(3) $\frac{2L}{\pi}$

(4) $\frac{L}{\pi^2}$

23. A shell of mass 20 kg at rest explodes into two fragments whose masses are in the ratio 2 : 3. The smaller fragment moves with a velocity of 6 m s^{-1}. The kinetic energy of the larger fragment is

(1) 96 J

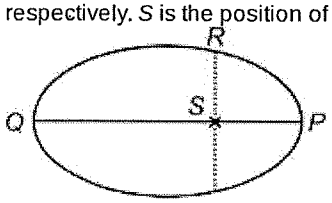
(2) 216 J

(3) 144 J

(4) 360 J

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24. The potential energy of a planet in an elliptical orbit about the sun at point P, Q and R are U_P, U_Q and U_R respectively. S is the position of sun. Then



(1) $U_P > U_R > U_Q$

(2) $U_Q > U_P > U_R$

(3) $U_Q > U_R > U_P$

(4) $U_Q = U_P = U_R$

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25. The energy required to raise a body of mass m from earth's surface to height of $3R$ (Where R is radius of earth) is

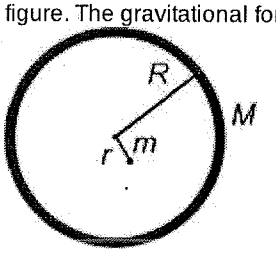
(1) mgR

(2) $\frac{3}{4}mgR$

(3) $\frac{2}{3}mgR$

(4) $\frac{5}{6}mgR$

26. A body of mass m is kept at distance r from the centre of a spherical shell of mass M and radius R as shown in figure. The gravitational force on the body is



(1) Zero

(2) $\frac{GMm}{r^2}$

(3) $\frac{GMm}{R^2}$

(4) $\frac{GMm}{(R-r)^2}$

27. Choose the correct option regarding Bulk Modulus of solid, liquid and gas.

(1) $B_{solid} > B_{gas} > B_{liquid}$

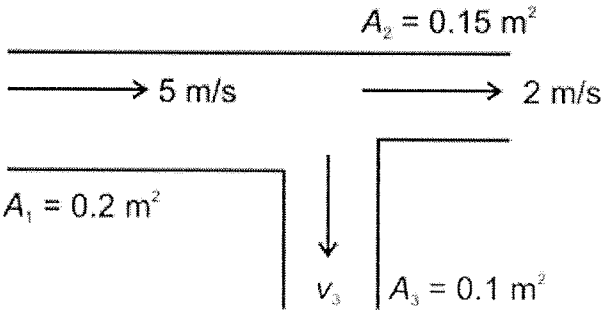
(2) $B_{gas} > B_{solid} > B_{liquid}$

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(3) $B_{solid} > B_{liquid} > B_{gas}$

(4) $B_{gas} > B_{liquid} > B_{solid}$

28. An ideal fluid is flowing through different cross-section as shown in figure, the velocity v_3 will be



(1) 7 m/s

(2) 3 m/s

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(3) 1 m/s

(4) 10 m/s

29. The period of revolution of a satellite around a planet of radius R is T. The period of revolution of another satellite around the same planet in an orbit of radius 2 R is

(1) $2T$

(2) $9T$

(3) T

(4) $2\sqrt{2}T$

30. Bernoulli's equation is valid for

(1) Viscous fluid

(2) Non-ideal fluid

(3) All types of fluid

(4) Ideal streamlined flow of fluid

31. The working of hydraulic lift is based on

(1) Torricelli's law

(2) Pascal's law

(3) Archimedes' principle

(4) Bernoulli's theorem

32. Young's modulus depends upon

(1) Stress applied on material

(2) Strain produced in material

(3) Temperature of material

(4) All of these

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33. The maximum load a wire can withstand without breaking is W. If radius of wire is double then the maximum load this wire can withstand without breaking is

(1) W

(2) $2W$

(3) $4W$

(4) $\frac{W}{2}$

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34. A wire is stretched by 0.2 mm by a certain weight. What will be the increase in length if same weight is hanged on other wire of same material and length but twice the radius of cross-section?

(1) 0.8 mm

(2) 0.1 mm

(3) 0.05 mm

(4) 0.025 mm

35. An object of specific gravity σ is hung from a massless string. The tension in string is *T*. The object is immersed in water so that one half of its volume is submerged. The new tension in the string is

(1) $\left(\frac{2\sigma+1}{2\sigma}\right)T$

(2) $\left(\frac{2\sigma-1}{2\sigma}\right)T$

(3) $\left(\frac{\sigma+1}{\sigma}\right)T$

(4) $\left(\frac{\sigma-1}{\sigma}\right)T$

SECTION-B

36. If 10th division of vernier scale coincides with 9th division of main scale in vernier calliper, then the least count of calliper is (take, 1 MSD = 1 mm)

(1) 0.02 mm

(2) 0.01 mm

(3) 0.03 mm

(4) 0.1 mm

37. A car moving with speed of 50 km/h can be stopped by brakes after atleast 6 m. If the same car is moving at speed of 100 km/h, the minimum stopping distance is

(1) 12 m

(2) 18 m

(3) 24 m

(4) 6 m

38. Two balls are thrown vertically up from same point on ground with same speed of 20 m/s at an interval of 2 s. At what height above ground they will hit each other?

(1) 20 m

(2) 15 m

(3) 10 m

(4) They will never hit

39. If initial velocity of an object is $(2\hat{j})$ m/s and it moves with constant acceleration $\left(\hat{i}+\hat{j}\right)$ m/s^2, then the velocity of particle at the end of 2 s is

(1) $\left(2\hat{i}+4\hat{j}\right)$ m/s

(2) $\left(2\hat{i}\right)$ m/s

(3) $\left(4\hat{i}\right)$ m/s

(4) $\left(2\hat{i}-4\hat{j}\right)$ m/s

40. Work done by a force, $\vec{F}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)$ N acting on a particle in displacing it from position $\vec{r}_1=\left(\hat{i}+\hat{j}+\hat{k}\right)$ m to $\vec{r}_2=\left(\hat{i}-\hat{j}+2\hat{k}\right)$ m is

(1) –3 J

(2) –1 J

(3) Zero

(4) 2 J

41. A uniform metre scale balances at 40 cm mark when weight of 20 g and 40 g are suspended at 10 cm and 20 cm mark respectively. The mass of the metre scale is

(1) 200 g

(2) 60 g

(3) 70 g

(4) 140 g

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42. Two bodies of masses 2 kg and 5 kg have position (1 m, 2 m, 1 m) and (3 m, 2 m, –1 m) respectively. The position vector of centre of mass is

(1) $\left(\frac{17}{7}\hat{i}+2\hat{j}-\frac{3}{7}\hat{k}\right)$ m

(2) $\left(17\hat{i}+14\hat{j}-3\hat{k}\right)$ m

(3) $\left(2\hat{i}+\frac{17}{7}\hat{j}-\frac{3}{7}\hat{k}\right)$ m

(4) $\left(14\hat{i}+17\hat{j}-3\hat{k}\right)$ m

43. Work done by friction is

(1) Always positive

(2) Always negative

(3) Always zero

(4) May be positive, negative or zero

44. A monkey is sitting on a tree limb. The limb exerts a normal force of 20 N and frictional force of 15 N. The magnitude of total force exerted by the limb on the monkey is

(1) 35 N

(2) 30 N

(3) 25 N

(4) 20 N

45. A train has to negotiate a curve of radius 400 m. By how much should the outer rail be raised w.r.t. inner rail for a speed of 72 km/h, if distance between the rails is 1.5 m (g = 10 m/s^2)

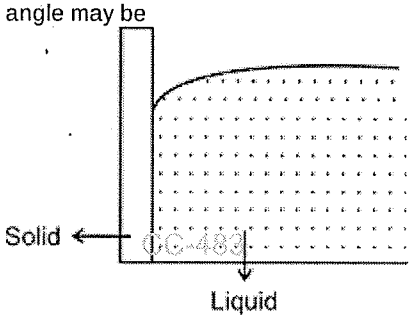
(1) 5 cm

(2) 7.5 cm

(3) 10 cm

(4) 15 cm

46. A liquid-solid contact is shown in the figure. The contact angle may be



(1) 0°

(2) 45°

(3) 90°

(4) 108°

47. An open vessel containing the liquid upto a height of 15 m. A small hole is made at height of 10 m from the base of the vessel, then the initial velocity of efflux is (g = 10 m/s^2)

(1) 1 m/s

(2) $10\sqrt{2}$ m/s

(3) 5 m/s

(4) 10 m/s

48. The excess of pressure inside the first soap bubble is $\sqrt{3}$ times that inside the second bubble. The ratio of volume of the first to that the second bubble is

(1) 1 : $3\sqrt{3}$

(2) 1 : $\sqrt{3}$

(3) $3\sqrt{3}$: 1

(4) $\sqrt{3}$: 1

49. The weight of a body on the surface of earth is 12.6 N. When it is raised to a height equal to half of the radius of earth its weight becomes

(1) 2.8 N

(2) 5.6 N

(3) 12.6 N

(4) 10.8 N

50. The excess pressure inside a soap bubble is equal to an oil column of height 3 mm. The total surface area of the soap bubble in contact with air will be (given, density of oil = 0.9 g/cm^3, surface tension of soap solution = 0.03 N/m and g = 10 ms^{-2})

(1) 4.96×10^{-4} m^2

(2) 1.24×10^{-4} m^2

(3) 3.72×10^{-4} m^2

(4) 1.03×10^{-4} m^2

CHEMISTRY

SECTION-A

51. A 66.2 watt bulb emits monochromatic light of wavelength 300 nm. The number of photons emitted by the bulb per second is

($h = 6.62 \times 10^{-34}$ Js)

(1) 10^{20}

(2) 10^{22}

(3) 10^{24}

(4) 10^{18}

52. Given below are two statements.

Statement I: In photoelectric effect, the number of electrons ejected depend upon the brightness of light.

Statement II: In photoelectric effect, the kinetic energy of the ejected electrons does not depend upon the brightness of light.

In the light of above statements, choose the correct answer from the options given below.

(1) Statement I is false but statement II is true

(2) Statement I is true but statement II is false

(3) Both statement I and statement II are true

(4) Both statement I and statement II are false

53. Match List-I with List-II.

	List I (Concentration terms)		List II (Units)
a.	Molarity	(i)	mole
b.	Mole fraction	(ii)	Unitless
c.	Molality	(iii)	mol L^{-1}
d.	Amount of substance	(iv)	mol kg^{-1}

Choose the correct answer from the options given below.

(1) a(iv), b(iii), c(ii), d(i)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iv), b(ii), c(i), d(iii)

54. Maximum number of atoms present in a plane in SbF_5 is

(1) 5

(2) 4

(3) 3

(4) 2

55. Match the following items in column I with column II and choose the correct option.

	Column I (quantum no. s)		Column II (orbital)
(a)	n = 2, l = 1	(i)	3*p*
(b)	n = 3, l = 2	(ii)	3*d*
(c)	n = 1, l = 0	(iii)	2*p*
(d)	n = 3, l = 1	(iv)	1*s*

(1) (a)(iii), (b)(i), (c)(iv), (d)(ii)

(2) (a)(iii), (b)(ii), (c)(iv), (d)(i)

(3) (a)(ii), (b)(i), (c)(iii), (d)(iv)

(4) (a)(i), (b)(ii), (c)(iii), (d)(iv)

56. Volume of 0.5 M NaOH required to prepare 0.2 M, 1 L NaOH solution is

(1) 300 ml

(2) 400 ml

(3) 200 ml

(4) 500 ml

57. Standard enthalpy of formation is zero for

(1) White phosphorous

(2) Diamond

(3) Hydrogen liquid

(4) Iodine liquid

58. An organic compound contains 40% carbon, 53.33% oxygen and rest hydrogen. The molecular formula of the compound can be

(1) C_2H_4O

(2) $C_2H_4O_2$

(3) $C_2H_6O_2$

(4) C_2H_6O

59. Incorrect statement about PCl_5 molecule is

(1) Hybridisation of P is sp^3d

(2) Shape of molecule is trigonal bipyramidal

(3) All the bond angles are not equivalent

(4) Three bonds are longer than the other two bonds

60. The general electronic configuration of *f*-block elements is

(1) $(n-2)\,f^{1\text{-}14}\,nd^{0-1}\,ns^2$

(2) $(n-2)\,f^{1\text{-}14}\,(n-1)\,d^{0-1}\,ns^2$

(3) $(n-2)\,f^{1\text{-}14}\,(n-1)\,d^{10}\,ns^2$

(4) $(n-2)\,f^{1\text{-}14}\,(n+1)\,d^2\,ns^2$

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61. Species which has square planar shape is

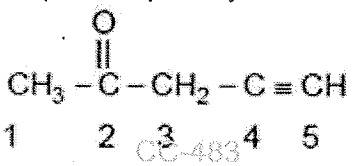
(1) $[PtCl_4]^{2-}$

(2) XeO_4

(3) SF_4

(4) CH_4

62. The hybridisation of C_1, C_2 and C_5 in the following compound respectively are



(1) sp^3, sp^2, sp

(2) sp^3, sp, sp^2

(3) sp, sp^2, sp^3

(4) sp^2, sp^2, sp

63. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$. What is the atomic number of the element, which is just below the above given element in the periodic table?

(1) 35

(2) 56

(3) 50

(4) 49

64. The correct order of increasing energy of orbitals for H-atom is

(1) $4s < 3d < 4p < 5s < 4d$

(2) $3d < 4s = 4p = 4d < 5s$

(3) $3d < 4s < 4p < 4d < 5s$

(4) $3d < 4s = 5s < 4p < 4d$

65. Maximum covalent character is present in

(1) NaF

(2) LiF

(3) KF

(4) CsF

66. Hydrogen bond is not observed in

(1) HCl

(2) H_2O

(3) HF

(4) NH_3

67. Which among the following hybrid orbitals has the maximum % s-character?

(1) sp

(2) sp^2

(3) sp^3

(4) sp^3d

68. What is the mass of the precipitate formed when 50 mL of 20.8% (w/V) solution of $BaCl_2$ is mixed with 50 mL of 14.2 % (w/V) Na_2SO_4 solution? (g atomic mass of barium = 137 g mol^{-1})

(1) 10 g

(2) 11.65 g

(3) 23.3 g

(4) 5.825 g

69. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): 0.031 has four significant figures.
Reason (R): Zeros preceding to first non-zero digit are significant.
In the light of above statements choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

70. Consider the following statements
Statement-I: Gay Lussac's Law of Gaseous Volumes : When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided that all gases are at same temperature and pressure.
Statement-II: The volumes of $H_2(g)$ and $O_2(g)$ which combine together to form water bear a simple ratio of 2 : 1
In the light of above statements, choose the correct answer from the options given below.

(1) Statement I is false but statement II is true

(2) Both statements I and II are true

(3) Both statements I and II are false

(4) Statement I is true but statement II is false

71. The enthalpy of combustion of methane at 25°C is –890 kJ mol^{-1}. The heat liberated when 4.8 g of methane is burnt in air is

(1) 445 kJ

(2) 267 kJ

(3) 890 kJ

(4) 178 kJ

72. 12.044×10^{22} molecules of glucose are present in 500 mL of its solution. The molarity of the solution is

(1) 0.01 M

(2) 0.4 M

(3) 0.04 M

(4) 0.2 M

73. The IUPAC official name of element having atomic number Z = 104 is

(1) Bohrium

(2) Rutherfordium

(3) Meitnerium

(4) Nobelium

74. In which of the following options, the order of arrangement does not agree with the variation of property indicated against it?

(1) Si < P < S < Cl (increasing electronegativity)

(2) Na > Li > Be (decreasing atomic radius)

(3) Rn < Xe < Kr < Ne (Increasing electron gain enthalpy)

(4) Na < Li < K < Rb (Increasing ionisation energy)

75. Consider the reactions:
$S_8(s) + 8\ O_2(g) \rightarrow 8\ SO_2(g)$, $\Delta H = -x$ kcal
$SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g)$, $\Delta H = -y$ kcal
The heat of formation of $SO_3(g)$ (in kcal) will be

(1) $-x - \frac{y}{8}$

(2) $\frac{-x}{8} - y$

(3) –8 x – 8 y

(4) –x – 8 y

76. The work done when 2 mole of a gas is expanded isothermally and reversibly at a temperature 27°C from 2 L to 4 L is

(1) –2.15 kJ

(2) –4.91 kJ

(3) –3.46 kJ

(4) –2.91 kJ

77. Angular momentum of the electron orbiting in the second orbit of hydrogen atom is

(1) $\frac{h}{\pi}$

(2) $\frac{2h}{\pi}$

(3) $\frac{h}{2\pi}$

(4) $\frac{h}{4\pi}$

78. Mass % of carbon in glucose ($C_6H_{12}O_6$) is

(1) 40%

(2) 50%

(3) 45%

(4) 35%

79. How many maximum number of electrons in an atom may have the quantum numbers n = 4 and $m_s = +\frac{1}{2}$?

(1) 32

(2) 16

(3) 8

(4) 12

80. Select the incorrect statement among the following.

(1) In the periodic table of elements, the atomic masses mentioned for different elements actually represent their average atomic masses

(2) One amu is defined as a mass exactly equal to one-twelfth the mass of one carbon-12 atom

(3) One mole is the amount of a substance that contains as many particles or entities as there are atoms in exactly 12 kg of the ^{12}C isotope

(4) The atomic mass of an element is expressed relative to ^{12}C isotope of carbon which has an exact value of 12 u

81. Which one of the following will have largest number of atoms?

(1) 16 g O_3

(2) 2 g H_2

(3) 4 g He

(4) 20 g Ca

82. Number of spectral lines obtained when electrons in a sample of hydrogen atom jump from n = 4 to n = 1 is

(1) 2

(2) 6

(3) 8

(4) 15

83. Select the species of highest negative electron gain enthalpy.

(1) F

(2) H

(3) O

(4) S

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84. For one mole of NaCl(s)

$\Delta_{lattice}H° = +788\ kJmol^{-1}$

$\Delta_{hyd}H° = -784\ kJmol^{-1}$

The enthalpy of solution is

(1) $-8\ kJ\ mol^{-1}$

(2) $-4\ kJ\ mol^{-1}$

(3) $-10\ kJ\ mol^{-1}$

(4) $+4\ kJ\ mol^{-1}$

85. The change in the hybridisation of B as result of the following reaction is

$BF_3 + NH_3 \rightarrow BF_3 : NH_3$

(1) sp^2 to sp^3

(2) sp^2 to sp

(3) sp to sp^3

(4) sp^3 to sp^2

SECTION-B

86. Molar heat capacity of aluminium is 25 $JK^{-1}\ mol^{-1}$. The heat necessary to raise the temperature of 54 g of aluminium (Atomic mass 27 g mol^{-1}) from 30°C to 50°C is

(1) 1.5 kJ

(2) 0.5 kJ

(3) 1.0 kJ

(4) 2.5 kJ

87. Reaction for which difference of ΔH and ΔU is maximum, is

(1) $TiI_4(g) \rightarrow Ti(s) + 2I_2(g)$

(2) $[Ni(CO)_4](g) \rightarrow Ni(s) + 4CO(g)$

(3) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

(4) $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$

88. Select the correct statement among the following

(1) If a rigid closed system has adiabatic boundaries then it must be isolated

(2) Entropy of a closed system is minimum at equilibrium

(3) Molar enthalpy is an extensive property

(4) Work done in a reversible adiabatic process is always zero

89. The bond order of 'Cl – O' bond in ClO_4^- is

(1) 1.5

(2) 2

(3) 1.75

(4) 1.25

90. Given below are two statements one is assertion(A) other is reason (R).

Assertion (A): NH_3 is more polar than NF_3.

Reason (R): NH_3 is pyramidal where as NF_3 is planar molecule.

In the light of above statements, choose the **correct** answer.

(1) Both Assertion and Reason are false statements

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(4) Assertion is true statement but Reason is false

91. The shape of I_3^+ ion is

(1) Trigonal planar

(2) Linear

(3) Tetrahedral

(4) V-shaped

92. Representative element among the following is

(1) Ti

(2) Tc

(3) Te

(4) Ta

93. Correct order of atomic radius of the given elements is

(1) Mg > Al > Si > Na

(2) Na > Mg > Al > Si

(3) Si > Al > Mg > Na

(4) Mg > Na > Al > Si

94. If energy required for the ionization of 0.2 g atom of Na is x kJ then the amount of energy required (in kJ) to ionize 2.3 g of Na atom is

(1) x

(2) $\frac{x}{2}$

(3) $\frac{x}{3}$

(4) 2x

95. The incorrect presentation for the given set of quantum numbers is

	n	l	m_l	s
(1)	4	2	–2	+1/2
(2)	3	2	–1	–1/2
(3)	2	1	–2	+1/2
(4)	5	3	0	+1/2

(1) (1)

(2) (2)

(3) (3)

(4) (4)

96. Maximum number of electrons for which n + l = 3, is

(1) 2

(2) 6

(3) 8

(4) 10

97. Number of unpaired electrons(s) in Cr^{3+} ion is

(1) 2

(2) 3

(3) 4

(4) 5

98. Number of Fe atoms in 22.4 g Haemoglobin if it contains 0.25% Fe (Atomic mass of Fe = 56) is

(1) 6.02×10^{21}

(2) 6.02×10^{20}

(3) 3.01×10^{20}

(4) 3.01×10^{21}

99. Mole fraction of solute in a 4.45 molal aqueous solution is

(1) 0.121

(2) 0.021

(3) 0.074

(4) 0.205

100. If 56 g N_2 and 64 g CH_4 are mixed with 80 g SO_3 gas, then mole fraction of CH_4 is

(1) 0.52

(2) 0.57

(3) 0.42

(4) 0.24

BOTANY

SECTION-A

101. Algal cell wall is made up of

(1) Cellulose, pectin, chitin

(2) Calcium pectate only

(3) Cellulose, galactans, mannans and calcium carbonate

(4) Cellulose, hemicelluloses, peptidoglycan

102. Cell membrane is mainly composed of

(1) Lipids and proteins

(2) Proteins and cholesterol

(3) Lipids and carbohydrates

(4) Proteins and nucleic acid

103. Important site for formation of glycoproteins and glycolipids in a typical eukaryotic cell is

(1) Ribosomes

(2) Lysosomes

(3) Endoplasmic reticulum

(4) Golgi apparatus

104. Select the **incorrect** match.

(1) Grana - Chloroplast

(2) Hydrolases - Lysosome

(3) Linear ds DNA - Mitochondria

(4) Centriole - Non-membrane bound organelle

105. Read the following statements and choose the **correct** option.
Statement A : In animal cells, steroidal hormones are synthesised by smooth endoplasmic reticulum.
Statement B : Hydrolases, digestive enzymes are present in Golgi bodies.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

106. Eukaryotic flagella and cilia are similar to each other as both
a. Have '9+2' organisation of microtubules
b. Are surrounded by membrane
c. Help in locomotion
d. Give cartwheel appearance with no microtubules in the centre.
The **correct** one(s) is/are

(1) Only a and b

(2) Only c

(3) Only a and d

(4) Only a, b and c

107. Identify the stage in which non-sister chromatids of chromosomes separate.

(1) Prophase I

(2) Anaphase I

(3) Telophase II

(4) Anaphase II

108. Which of the following event is related to prophase?

(1) Centriole duplication

(2) Chromatin condensation

(3) Reappearance of Golgi complex

(4) Reappearance of nuclear membrane

109. In which of the following stages, terminalisation of chiasmata occurs?

(1) Leptotene

(2) Diakinesis

(3) Pachytene

(4) Zygotene

110. Which of the given statement is **not** related to cytokinesis of a plant cell?

(1) New cell wall formation begins with the formation of a precursor called cell-plate

(2) Distribution of plastids in two daughter cells

(3) Wall formation starts in the centre

(4) Formation of a furrow in the plasma membrane

111. What will be the amount of DNA and number of chromosomes in a daughter cell produced after meiosis II if the meiocyte contains 40 pg of DNA and 20 chromosomes in G_1 phase?

(1) 10 pg, 10 chromosomes

(2) 20 pg, 10 chromosomes

(3) 40 pg, 20 chromosomes

(4) 10 pg, 20 chromosomes

112. Find the **correct** feature of interkinesis during cell division.

(1) Replication of DNA

(2) Formation of chiasmata

(3) Duplication of centriole in animal cell

(4) Formation of synaptonemal complex

113. Arrange the following in **correct** sequence that happen during cell division and select the option accordingly.
a. Disappearing of mitotic spindle.
b. Migration of daughter chromosomes.
c. Beginning of centrioles movement towards opposite poles.
d. Attaching of spindle microtubules to the kinetochore.

(1) d, c, b, a

(2) c, d, b, a

(3) b, c, a, d

(4) d, c, a, b

114. Read the following statements of Assertion (A) and Reason (R) and choose the **correct** option
Assertion (A) : White spots seen on mustard leaves are due to a parasitic fungus.
Reason (R) : Most fungi are heterotrophic and absorb soluble organic matter from dead substrates.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

115. Causal agent of malaria is a/an

(1) Amoeboid protozoan

(2) Ciliated protozoan

(3) Sporozoan

(4) Flagellated protozoan

116. Organisms placed in the kingdom Fungi have all of the following features, **except**

(1) Chitinous cell wall

(2) Heterotrophic mode of nutrition

(3) Lack of nuclear membrane

(4) Eukaryotic cell type

117. Vast majority of bacteria are

(1) Phototrophic

(2) Chemosynthetic

(3) Autotrophic

(4) Heterotrophic

118. No virus contains

(1) Proteinaceous capsid

(2) Envelope

(3) Both RNA & DNA as their genetic material

(4) Double stranded DNA

119. A feature which is **not** common between sac fungi and imperfect fungi is

(1) Presence of septate and branched mycelium

(2) Formation of endogenous sexual spores

(3) Asexual reproduction by conidia

(4) Parasitic and saprophytic mode of nutrition

120. Cyanobacteria

(1) Are filamentous heterotrophic bacteria

(2) Lack chlorophyll-a pigment

(3) Show photosynthesis

(4) Have membrane bound cell organelles

121. Which among the following pairs, does not have taxa at the same hierarchial level?

(1) Hominidae, poaceae

(2) Diptera, polymoniales

(3) Monocotyledonae, mammalia

(4) Chordata, insecta

122. Among kingdom, phylum, class, order and family how many taxonomic categories are common for *Felis* and *Canis*?

(1) Two

(2) Three

(3) Four

(4) Five

123. Which one of the following is **incorrect** w.r.t. rules of binomial nomenclature.

(1) Each organism is given two names consisting of two words

(2) The biological names are generally taken from Latin language irrespective of their origin

(3) The scientific name is printed in italics

(4) The first word denoting the genus starts with capital letter while the specific epithet starts with a small letter

124. In which of the following organisms, growth and reproduction are mutually inclusive events?

(1) Humans

(2) Mule

(3) *Amoeba*

(4) Honey bee

125. Which of the given is highest category in taxonomic hierarchy?

(1) Phylum

(2) Kingdom

(3) Species

(4) Order

126. Which of the following plants does **not** belong to the order Polymoniales?

(1) Potato

(2) *Petunia*

(3) *Datura*

(4) Wheat

127. The characteristic feature of both prophase II and diakinesis is

(1) Formation of centrosome

(2) Terminalisation of chiasmata

(3) Disappearance of nucleolus and nuclear membrane

(4) Reappearance of nuclear membrane

128. Viroids

(1) Are infectious proteins

(2) Have capsid as outer covering

(3) Were discovered by T.O. Diener

(4) Are viruses with ssDNA

129. In bean flower, the largest posterior petal is known as

(1) Keel

(2) Wing

(3) Standard

(4) Bract

130. A small pore above the scar found on seed coat is

(1) Micropyle

(2) Hilum

(3) Plumule

(4) Tegmen

131. Stilt roots

(1) Are modified to get oxygen

(2) Support the stem in sugarcane

(3) Grow vertically upwards

(4) Are modified for storage of food

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132. Flower has bilateral symmetry in

(1) Mustard

(2) *Datura*

(3) Bean

(4) Chilli

133. Which of the following plants does **not** have the modified storage tap root?

(1) Carrot

(2) Sweet potato

(3) Radish

(4) Turnip

134. Syncarpous gynoecium is/are found in

(a) Mustard
(b) Tomato
(c) Lotus
(d) Rose

(1) Only (c)

(2) Only (a) and (b)

(3) Only (c) and (d)

(4) Only (a), (b) and (d)

135. Select the **incorrect** statement.

(1) Staminodes are sterile stamens

(2) Prop roots in banyan tree are adventitious roots

(3) Radicle in maize seed is enclosed in coleorhiza

(4) Mesocarp and endocarp in mango are fleshy and edible

SECTION-B

136. Resistance to antibiotics in bacteria is mainly due to

(1) Genomic DNA

(2) Cellulosic cell wall

(3) Small circular DNA called plasmid

(4) Plasma membrane

137. Bacterial cells may be motile or non-motile. If motile, which of the following structures help in their motility?

(1) Pili

(2) Fimbriae

(3) Flagella

(4) Cilia

138. Select the **incorrect** statement about glycocalyx in bacteria.

(1) Can be present in the form of a loose sheath

(2) It is the outermost layer of cell envelope

(3) Differs in composition and thickness among different bacteria

(4) Is always hard and thick

139. Nuclear envelope of a nucleus

(1) Separates nucleoplasm from the cytoplasm.

(2) Is present in *Mycoplasma*

(3) Is smooth and never have ribosomes on it

(4) Contains nuclear pore that controls one way movement of substances *i.e.*, only from nucleus to cytoplasm.

140. Read the statement and select the **correct** option.

A: The cis and the trans faces of golgi apparatus are entirely different but are not interconnected.
B: Golgi apparatus remains in close association with E.R.
C: Cisternae in golgi apparatus are stacked parallel to each other

(1) Only statement A is incorrect.

(2) Statements A and C are correct.

(3) All statements A, B and C are correct.

(4) Statements A and B are incorrect.

141. During which phase of cell cycle, genetic material becomes double?

(1) Gap_2 phase

(2) Synthesis phase

(3) Gap_1 phase

(4) Quiescent phase

142. Bivalent formation takes place in

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

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143. Consider the following statements (A to C) and select the **correct** option stating which ones are **True (T)** and which ones are **False (F)**.
A. Ostrich egg is the largest isolated single cell.
B. The shape of the cells may vary with the function they perform.
C. Plasmodesmata glue the different neighbouring cells together.

	A	B	C
(1)	T	F	T
(2)	T	T	F
(3)	T	F	F
(4)	F	T	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

144. Aristotle classified the animals on the basis of
(1) Morphological characters
(2) Phylogenetic relationships
(3) Natural affinities among them
(4) Absence or presence of red blood

145. Heterotrophic bacteria are helpful in all, **except**
(1) Decomposing organic material
(2) Production of antibiotics
(3) Fixing nitrogen in legumes
(4) Recycling nutrients by oxidising various inorganic substances

146. Important feature which is **not** included in modern taxonomic studies, is
(1) External and internal structure of organism
(2) Developmental processes
(3) Ecological information of organisms
(4) Evolutionary relationships between organisms

147. Twisted type of aestivation in corolla is found in the flowers of
(1) *Calotropis*
(2) China rose
(3) Pea
(4) *Cassia*

148. The placenta develops at the base of the ovary in
(1) Marigold
(2) *Dianthus*
(3) *Argemone*
(4) China rose

149. Read the statements of assertion and reason and choose the correct option.
Assertion (A) : Flower is a modified shoot, meant for sexual reproduction.
Reason (R) : Floral characteristics form the basis of classification and identification of flowering plants.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

150. A compound leaf differs from a simple leaf as the former
(1) Bears axillary buds in axil of its leaflets
(2) Has reticulate venation
(3) Has a number of leaflets as incisions of the lamina reach upto the midrib
(4) Bears a bud in the axil of the petiole

ZOOLOGY

SECTION-A

151. The type of epithelium present in the walls of blood vessels and air sacs of lungs is
(1) Squamous epithelium
(2) Cuboidal epithelium
(3) Columnar epithelium
(4) Ciliated epithelium

152. Ciliated epithelium lining the bronchioles helps in
(1) Exchange of gases
(2) Reabsorption of water
(3) Filtration of blood
(4) Movement of mucus and trapped particles in a specific direction

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153. The epithelium that plays limited role in secretion and absorption is

(1) Simple squamous epithelium

(2) Simple cuboidal epithelium

(3) Simple columnar epithelium

(4) Compound epithelium

154. Select the tissue that exerts the greatest control over the body's responsiveness to changing conditions.

(1) Connective tissue

(2) Muscular tissue

(3) Nervous tissue

(4) Epithelial tissue

155. How many of the given features in the box below is/are present in smooth muscle fibres?

Fusiform appearance, Uninucleated, Multinucleated, Branched appearance, Presence of striations

Select the correct option.

(1) One

(2) Two

(3) Three

(4) Four

156. Enzymes catalysing removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds are categorised under _____ of enzymes.
Choose the correct option to fill in the blank.

(1) Class II

(2) Class III

(3) Class IV

(4) Class VI

157. Biomolecule that is **not** obtained in acid soluble fraction after chemical analysis of tissue is

(1) Glucose

(2) Alanine

(3) Cholesterol

(4) Ribose

158. Read the given statements and choose the **correct** option.
Statement I : In solutions of different pH, the structure of amino acids changes.
Statement II : An amino acid may exist as zwitterionic form, anionic form or cationic form depending on the pH of the solution.

(1) Only statement I is correct

(2) Only statement II is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

159. Catalytic activity is lost when the_____ is removed from the enzyme
Select the option which fills the blank **correctly**.

(1) Cofactor

(2) Competitive inhibitor

(3) Product

(4) Non-competitive inhibtor

160. Triglycerides are formed by esterification of X of glycerol with Y of fatty acids. Identify X and Y from the options given below to complete the statement.

(1) X = 3 molecules, Y = 1 molecule

(2) X = 1 molecule, Y = 3 molecules

(3) X = 1 molecule, Y = 1 molecule

(4) X = 3 molecules, Y = 3 molecules

161. Conducting part of human respiratory system does **not** include

(1) Trachea

(2) Pharynx

(3) Bronchi

(4) Alveoli

162. Which of the following conditions lower the affinity of haemoglobin for oxygen?

(1) High pH, high pCO_2 and low temperature

(2) Low pH, high pCO_2 and high temperature

(3) Low pH, low pCO_2 and low temperature

(4) High pH, low pCO_2 and high temperature

163. Functional residual capacity can be represented by

(1) ERV + RV

(2) IRV + TV

(3) VC + RV

(4) TV + ERV

164. Branchial respiration is the term used for respiration by

(1) Lungs

(2) Gills

(3) Trachea

(4) Skin

165. In systemic arteries, partial pressure of oxygen is

(1) 40 mm Hg

(2) 95 mm Hg

(3) 45 mm Hg

(4) 104 mm Hg

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166. Under normal conditions, every 1000 mL of deoxygenated blood delivers approximately ___ mL of CO_2 to the alveoli.
Select the correct option to fill in the blank.

(1) 4

(2) 5

(3) 40

(4) 50

167. The **correct** pair of agranulocytes among the following is

(1) Monocytes and basophils

(2) Monocytes and lymphocytes

(3) Neutrophils and lymphocytes

(4) Eosinophils and lymphocytes

168. The first heart sound occurs due to
(a) Closure of tricuspid valve
(b) Opening of semilunar valve
(c) Closure of bicuspid valve
(d) Closure of semilunar valves.
Choose the **correct** option.

(1) (a) and (c)

(2) (b) and (c)

(3) (b) only

(4) (a) and (d)

169. A state of heart when it is not pumping blood effectively to meet the needs of the body is called

(1) Angina pectoris

(2) Atherosclerosis

(3) Cardiac arrest

(4) Heart failure

170. Tunica media of blood vessels is made up of

(1) Connective tissue with collagen fibre

(2) Cuboidal epithelium

(3) Columnar epithelium

(4) Smooth muscle and elastic fibres

171. The average life span of a human RBC is A and the organ considered graveyard of erythrocytes is B.
Choose the option that fills the blanks **correctly**.

(1) 10 days; spleen

(2) 120 days; liver

(3) 120 days; spleen

(4) 180 days; liver

172. In humans, specialised cardiac musculature responsible for generating the maximum number of action potentials is present in

(1) Right upper corner of the right atrium

(2) Left upper corner of the left atrium

(3) Lower left corner of the right atrium

(4) Right upper corner of the right ventricle

173. In humans, kidneys are situated between the levels of

(1) Third lumbar and fifth lumbar vertebra

(2) Last thoracic and third lumbar vertebra

(3) Last lumbar and third sacral vertebra

(4) Last cervical and third thoracic vertebra

174. The cortex extends in between the medullary pyramids as renal columns called

(1) Hilum

(2) Calyces

(3) Columns of Bertini

(4) Capsule

175. On an average, A mL of blood is filtered by the kidneys per minute which constitutes roughly B of the blood pumped out by each ventricle of the heart in a minute.
Choose the option which **correctly** fills the blanks A and B respectively w.r.t. an adult human under normal physiological conditions.

(1) 125, 1/2

(2) 1100 – 1200, $1/5^{th}$

(3) 700 – 800, $1/4^{th}$

(4) 1100 – 1200, $3/4^{th}$

176. Juxta medullary nephrons differ from cortical nephrons in having

(1) Bowman's capsule

(2) PCT

(3) Longer loop of Henle

(4) Highly reduced vasa recta

177. Read the given statements and choose the **correct** option.
Statement A: The process of release of urine is called micturition reflex.
Statement B: The CNS passes on motor messages to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter causing the release of urine.

(1) Both statements are incorrect

(2) Both statements are correct

(3) Only statement A is correct

(4) Only statement B is correct

178. Which of the following is not a part of renal tubule?

(1) PCT

(2) DCT

(3) Loop of Henle

(4) Glomerulus

179. In an adult human, which category of vertebrae are found in fused form condition?

(1) Sacral and coccygeal

(2) Cervical and thoracic

(3) Thoracic and lumbar

(4) Cervical and coccygeal

180. In a sarcomere, I-band has

(1) Only myosin filaments

(2) Only actin filaments

(3) Both actin and myosin filaments

(4) Neither actin nor myosin filaments

181. In a healthy person, the red muscle fibres appear red due to presence of abundant amount of

(1) Carboxyhemoglobin

(2) Carbaminohemoglobin

(3) Myoglobin

(4) Erythropoietin

182. In humans, A is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh rib.
Select the option that fills the above blank correctly.

(1) Coxal bone

(2) Clavicle

(3) Scapula

(4) Sternum

183. The joint between carpals is

(1) Pivot joint

(2) Gliding joint

(3) Saddle joint

(4) Fibrous joint

184. During maximal contraction of a skeletal muscle
(A) The Z line is pulled inwards thereby causing a shortening of the sarcomere.
(B) The I band and A band get reduced
(C) H zone disappears
Select the **correct** statements.

(1) A and B

(2) B and C

(3) A and C

(4) A, B and C

185. The _____ and _____ pairs of ribs are floating ribs because one end of the rib is attached to _____ and the other end is free. Fill in the respective blanks with correct answer.

(1) 11th, 12th, vertebral column

(2) 11th, 12th, sternum

(3) 11th, 14th, sternum

(4) 10th, 11th, vertebral column

SECTION-B

186. Which of the following animals possess 3-chambered heart?

(1) Crocodiles

(2) Fishes

(3) Amphibians

(4) Birds

187. Choose the blood cells which normally form 6-8% of all WBCs and are phagocytic in nature.

(1) Basophils

(2) Neutrophils

(3) Monocytes

(4) Lymphocytes

188. Complete the following reaction

$$\text{Prothrombin} \xrightarrow{A} \text{Thrombin}$$

$$B \xrightarrow{\text{Thrombin}} \text{Fibrin}$$

Choose the **correct** option w.r.t. A and B.

(1) A – Calcium B – Thrombokinase

(2) A – Fibrinogen B – Thromboplastin

(3) A – Thrombokinase B – Fibrinogen

(4) A – Fibrinogen B – Thrombokinase

189. Neural signals through the sympathetic nerve fibres can increase all of the following, **except**

(1) Heart rate

(2) Duration of cardiac cycle

(3) Cardiac output

(4) Speed of conduction of action potential

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190. **Assertion (A)** : With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first, the reaction ultimately reaches a maximum velocity (V_{max}) which is not exceeded by any further rise in concentration of the substrate.
Reason: (R) The enzyme molecules are fewer than the substrate molecules and after saturation of the molecules, there are no free enzyme molecules to bind with the additional substrate molecules.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true and (R) is false

191. Select the option which includes all aromatic amino acids.

(1) Valine, Glycine, Glutamic acid

(2) Tyrosine, Phenylalanine, Lysine

(3) Tryptophan, Tyrosine, Serine

(4) Phenylalanine, Tyrosine, Tryptophan

192. Which of the following statement is **incorrect** w.r.t. proteins?

(1) Proteins are homopolymers of amino acids

(2) Dietary proteins are source of essential amino acids

(3) Our body can synthesize non-essential amino acids

(4) RuBisCO is the most abundant protein in the whole of biosphere

193. A specialized centre which can moderate the functions of the respiratory rhythm centre is located in

(1) Corpora quadrigemina

(2) Midbrain

(3) Pons

(4) Cerebellum

194. Contraction of diaphragm during inspiration leads to

(1) Increase in the volume of thoracic chamber in the dorso-ventral axis

(2) Increase in the volume of thoracic chamber as well as pulmonary cavity in the antero-posterior axis

(3) Decrease in the volume of pulmonary cavity but increase in that of thoracic chamber

(4) No change in the volume of thoracic chamber

195. All of the following are supported by incomplete cartilaginous rings **except**

(1) Secondary bronchi

(2) Tertiary bronchi

(3) Terminal bronchioles

(4) Initial bronchioles

196. Select the **incorrect** pair w.r.t. excretory organs found in organisms.

(1) Protonephridia - *Planaria*

(2) Green glands - Prawn

(3) Nephridia - Earthworm

(4) Flame cells - Crustaceans

197. Read the following statements and choose the **correct** option.
Statement A: Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes like urea and uric acid for conservation of water.
Statement B: Ammonia is the least toxic form and requires large amounts of water for its elimination.

(1) Statement A is correct

(2) Both statements are correct

(3) Both statements are incorrect

(4) Statement B is correct

198. Bones have___A___and___B___ground susbstance rich in collagen fibres which give bone its___C___
Choose the option that fill the blanks correctly

	A	B	C
(1)	Hard	Pliable	Strength
(2)	Solid	Pliable	Flexibility
(3)	Solid	Non-Pliable	Stretchability
(4)	Hard	Non-Pliable	Strength

(1) (1)

(2) (2)

(3) (3)

(4) (4)

199. Ligaments are structures which connect

(1) Skeletal muscles to bones

(2) Bone to bone

(3) Non-striated muscles to bones

(4) Cardiac muscles to bones

200. Secretions of exocrine glands **exclude**

(1) Saliva

(2) Insulin

(3) Ear wax

(4) Milk

20-11-2024

Code-D

Phase-4_RM

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NCERT BOOSTER TEST SERIES

(*for* NEET-2025)

Test - I

Answers

1. (3)	41. (4)	81. (2)	121. (4)	161. (4)
2. (2)	42. (1)	82. (2)	122. (3)	162. (2)
3. (2)	43. (4)	83. (1)	123. (1)	163. (1)
4. (3)	44. (3)	84. (4)	124. (3)	164. (2)
5. (3)	45. (4)	85. (1)	125. (2)	165. (2)
6. (2)	46. (4)	86. (3)	126. (4)	166. (3)
7. (1)	47. (4)	87. (2)	127. (3)	167. (2)
8. (2)	48. (1)	88. (1)	128. (3)	168. (1)
9. (1)	49. (2)	89. (3)	129. (3)	169. (4)
10. (1)	50. (1)	90. (4)	130. (1)	170. (4)
11. (2)	51. (1)	91. (4)	131. (2)	171. (3)
12. (4)	52. (3)	92. (3)	132. (3)	172. (1)
13. (1)	53. (3)	93. (2)	133. (2)	173. (2)
14. (4)	54. (2)	94. (2)	134. (2)	174. (3)
15. (4)	55. (2)	95. (3)	135. (4)	175. (2)
16. (3)	56. (2)	96. (3)	136. (3)	176. (3)
17. (1)	57. (1)	97. (2)	137. (3)	177. (4)
18. (3)	58. (2)	98. (2)	138. (4)	178. (4)
19. (3)	59. (4)	99. (3)	139. (1)	179. (1)
20. (2)	60. (2)	100. (2)	140. (1)	180. (2)
21. (4)	61. (1)	101. (3)	141. (2)	181. (3)
22. (1)	62. (1)	102. (1)	142. (2)	182. (3)
23. (1)	63. (4)	103. (4)	143. (2)	183. (2)
24. (3)	64. (2)	104. (3)	144. (4)	184. (3)
25. (2)	65. (2)	105. (1)	145. (4)	185. (1)
26. (1)	66. (1)	106. (4)	146. (4)	186. (3)
27. (3)	67. (1)	107. (2)	147. (2)	187. (3)
28. (1)	68. (2)	108. (2)	148. (1)	188. (3)
29. (4)	69. (4)	109. (2)	149. (2)	189. (2)
30. (4)	70. (2)	110. (4)	150. (3)	190. (1)
31. (2)	71. (2)	111. (2)	151. (1)	191. (4)
32. (3)	72. (2)	112. (3)	152. (4)	192. (1)
33. (3)	73. (2)	113. (2)	153. (4)	193. (3)
34. (3)	74. (4)	114. (2)	154. (3)	194. (2)
35. (2)	75. (2)	115. (3)	155. (2)	195. (3)
36. (4)	76. (3)	116. (3)	156. (3)	196. (4)
37. (3)	77. (1)	117. (4)	157. (3)	197. (1)
38. (2)	78. (1)	118. (3)	158. (3)	198. (4)
39. (1)	79. (2)	119. (2)	159. (1)	199. (2)
40. (2)	80. (3)	120. (3)	160. (2)	200. (2)

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